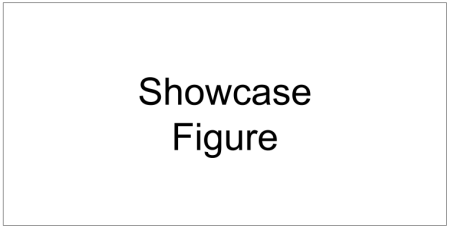
LMU LUDWIG-MAXIMILIANS-UNIVERSITÄT MÜNCHEN

Understanding Trajectron++: A Per-Trajectory Performance Analysis

Simon Drauz and Zoe Matrullo

May 17, 2026



Outline

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Trajectory Prediction for Autonomous Driving

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- Challenging due to **complex agent interactions** and influence of **static context**
- Essential for **safe path planning** and **collision avoidance** in autonomous vehicles

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- Handles *heterogeneous* agent types jointly (pedestrian, bicycle, car) and integrates HD map context

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Prediction metrics:

`ml_ade`	most-likely ADE ← **used**
`ml_fde`	most-likely FDE
`min_ade_5`	best-of-5 ADE
`nll_mean`	negative log-likelihood

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2. What are the *success and failure modes* that explain model performance?
3. How do model settings (history window, prediction horizon, attention radius) affect performance?

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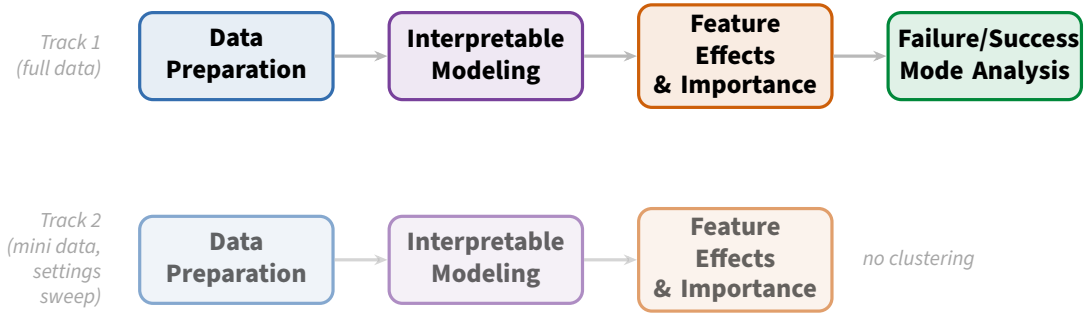

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- `data_idx`: stable trajdata identifier, used to join evaluation output with computed trajectory & scene characteristics

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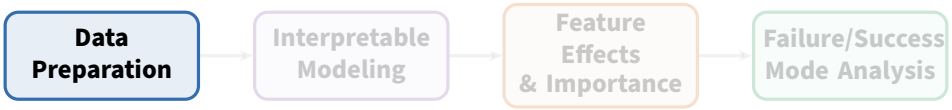

Feature Engineering — Trajectory & Scene Characteristics

Agent motion features

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- Speed: mean, max, std
- Acceleration: mean, max
- Jerk: mean, max
- Heading change per second
- Min. neighbour distance
- Collision indicator

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$\Rightarrow$ **18 candidate features** joined with Trajectron++ evaluation output via `data_idx`

Target Distribution & Log Transformation

- `ml_ade` is strongly *right-skewed* (skewness $= 2.59$)

[figure placeholder: before/after distribution]

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- Models are trained on log-scale target, evaluated on original scale

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Approach 1: MI elbow + VIF

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Motion features only; all highly significant ($p \ll 0.01$).

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Retains scene-level features alongside motion features.

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VIF only was used for the main analysis — it retains richer context without discarding potentially informative scene features.

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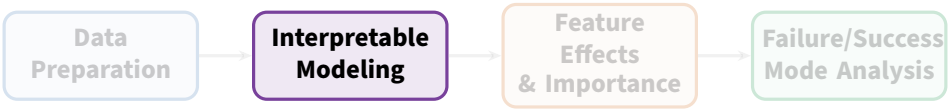

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- This turns ADE into actionable understanding of model behaviour

GAM and XGBoost

We tested two interpretable modeling approaches — either can be used for the analysis.

GAM (Generalized Additive Model) **XGBoost**

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- Post-hoc interpretation via SHAP values

Interpreting the Models: Additive Effects & SHAP Values

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- *Local:* why was this trajectory's ADE high?
- *Global:* mean $|\text{SHAP}|$ gives feature importance and direction

Validating the Tuning Procedure — Nested Resampling

- **Problem:** hyperparameter tuning risks overfitting to a particular data split
- **Solution:** nested cross-validation
  - Outer folds: evaluate generalization of the full tuning procedure
  - Inner folds: tune hyperparameters on each outer-fold training set via cross-validation
- OOF evaluation: inspect out-of-fold predictions for instability, outlier folds, bias
- **Conclusion:** tuning generalises stably $\rightarrow$ proceed to full-data tuning + refit

Final models are refit on *all* data with optimal hyperparameters — no held-out test set, because the goal is understanding, not unbiased prediction.

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Three candidates:

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- **LinearGAM** — fitted on raw `ml_ade`
- **LinearGAM (log)** — fitted on $\log(\texttt{ml\_ade})$, evaluated on original scale

Variant	RMSE	R^2	MAE
LinearGAM	**0.440**	**0.515**	0.268
LinearGAM (log)	0.444	0.506	**0.255**
GammaGAM	0.475	0.434	0.277

Outer-fold means, 3-fold nested CV.
LinearGAM selected.

Selection criterion: lowest mean outer-fold RMSE (original scale).

GAM Variant Selection

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- **LinearGAM** — fitted on raw `ml_ade`
- **LinearGAM (log)** — fitted on log(`ml_ade`), evaluated on original scale
- **GammaGAM** — Gamma distribution with log link, suited for positive right-skewed targets

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Final Model Performance (OOF)

GAM (LinearGAM)

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- Additive smooth terms, one per feature

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Metric	OOF value
RMSE	**0.387**
R^2	**0.624**
MAE	**0.213**

XGBoost outperforms GAM on all metrics. Both models are used as a cross-check: the feature-effect narratives should agree across approaches.

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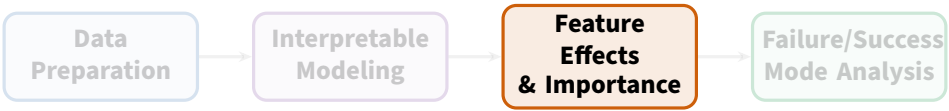

Global Feature Importance

XGBoost — mean $|\mathrm{SHAP}|$

Feature	**Mean** $\lvert\mathrm{SHAP}\rvert$
`std_speed`	**0.129**
`heading_change_per_sec`	0.070
`min_neighbor_distance`	0.027
`mean_acceleration`	0.027
`scene_density_VEHICLE`	0.020
`scene_num_VEHICLE`	0.016

GAM — smooth-term significance

Feature	$p < 0.05$**?**
`heading_change_per_sec`	✓
`mean_acceleration`	✓
`min_neighbor_distance`	✓
`std_speed`	✓
`scene_num_VEHICLE`	✓
`scene_density_VEHICLE`	×

Both models agree: speed variation (`std_speed`) and turning (`heading_change_per_sec`) are the dominant drivers of prediction error. Scene density is consistently least important.

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Scene & proximity features

- `min_neighbor_distance`: mild positive effect — isolated agents are marginally harder to predict
- `scene_num_VEHICLE`: small positive effect; crowded vehicle context increases error slightly
- `scene_density_VEHICLE`: not significant in GAM; weakest SHAP contribution in XGBoost

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$\Rightarrow$ Trajectron++ struggles most with **fast, erratic, turning pedestrians**; scene context is a secondary factor.

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- Global picture consistent with GAM: `std_speed` contributes most strongly
- `min_neighbor_distance` shows a non-monotonic pattern — very close *and* very distant neighbors raise error

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- `heading_change_per_sec`: positive, non-linear — effect levels off at high turning rates
- Confidence bands narrow in high-density regions

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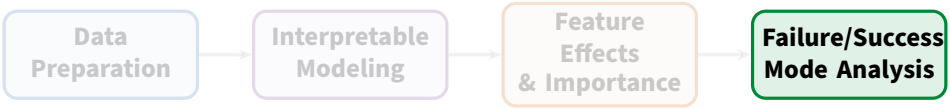

Performance Regime Methodology

1. Divide trajectories into difficulty groups based on `ml_ade` (easy / medium / hard)
2. Run clustering on trajectory + scene characteristics per group
   - Option A: original feature space
   - Option B: UMAP-reduced space (dimensions chosen via trustworthiness score + elbow rule)
3. Assess clustering quality with DBCV score
4. Export identified clusters (performance regimes)

Clustering Results

Performance group sizes (VIF-only run)

Group	Boundary	N	%
Easy	< 0.39	15 348	59%
Medium	0.39–1.20	8 094	31%
Hard	> 1.20	2 680	10%

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Cluster quality by group

Group	DBCV	Noise %
Easy	**0.411**	30%
Medium	0.397	40%
Hard	0.091	75%

Easy group has the clearest substructure.
Hard is mostly a continuous high-error regime.

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Easy regime

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Hard regime

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- Dominated by high `std_speed` ($+2.0\sigma$ global), turning, and acceleration
- No sharp sub-clusters — hard is a **continuous high-variation regime**; a small extreme outlier group has `std_speed` $\approx +4.8\sigma$

Takeaway: Trajectron++ struggles with erratic, fast-turning pedestrians. Slow or near-stationary pedestrians are easy regardless of scene complexity.

Outline

Settings Sweep Setup

- Dataset: nuScenes mini (full dataset too expensive for repeated runs)
- Cartesian sweep over:
  - `history_sec`
  - `prediction_sec`
  - `attention_radius_m`
- Pooled model: model setting variables + trajectory/scene features as control variables
- Feature normalization: setting-agnostic versions to avoid mechanical confounding (e.g. `mean_path_length_per_second` instead of `path_length`)

Effect of Model Settings

XGBoost — mean $|\mathrm{SHAP}|$ (pooled model)

Feature	Mean $\lvert\mathrm{SHAP}\rvert$
`prediction_sec`	**0.178**
`attention_radius_m`	0.098
`std_speed`	0.093
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`history_sec`	0.022

All 8 features significant in GAM ($p \ll 0.01$).

Key findings

- `prediction_sec`: largest effect — longer horizon means harder forecasting (ADE grows with time)

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- Trajectory features (`std_speed`, `max_speed`) retain strong explanatory power even after controlling for settings

Outline

Summary of Findings

1. **RQ1 — Which characteristics drive prediction error?**
   - `std_speed` (speed variation) and `heading_change_per_sec` (turning) are the dominant drivers in both GAM and XGBoost
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3. **RQ3 — How do model settings affect performance?**
   - Prediction horizon (`prediction_sec`) has the largest effect — longer horizon raises error strongly
   - Attention radius (`attention_radius_m`) is second; history window (`history_sec`) has minimal impact

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- Meta-models explain Trajectron++ predictions, not ground-truth difficulty — effects are model-specific
- Hard-regime clustering is weak (75% noise); discrete failure-mode identification remains an open challenge
- Settings sweep uses nuScenes *mini* — results may not transfer to full-scale training

Future Work

- Extend pipeline to vehicle and cyclist trajectories
- Apply to other trajectory prediction models and datasets via the trajdata interface
- Explore higher-capacity meta-models to better isolate failure modes in the hard regime